RSI (Rapid Sequence Intubation)

- RSI is indicated for a patient with a GCS less than or equal to 10, and
  - Clinically significant compromise of airway or ventilation, or
  - A clinical trajectory requiring RSI.
- Use the PODCASTS acronym to consider factors contributing to the balance of risk prior to proceeding with RSI.

P Predicted difficulty of intubation.

O Oxygenation and ventilation (severity of impairment).

D Distance (time) to hospital.

C Comorbidities that limit life expectancy.

A Age, particularly those aged less than 5 years, or greater than 75 years.

S Shock.

T Trajectory of clinical course.

S Special circumstances.

RSI procedure

- Ensure adequate preparation and resuscitation before commencing RSI.
- Utilise the RSI checklist.
- Administer 2 mcg/kg of fentanyl IV (up to a maximum of 200 mcg) immediately prior to ketamine if the patient has:
  - Traumatic brain injury, or
  - Suspected intracranial pathology, or
  - Hypertension (SBP > 150 mmHg in an adult).
  - Withhold fentanyl if the above clinical indications are not present, or there are signs of shock.
- Administer 2 mg/kg of ketamine IV (up to a maximum of 200 mg) for most patients.
  - Consider reducing the dose to 1 mg/kg if:
    - There are signs of shock, or
    - Sedative medicines have already been administered or taken.

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- Administer neuromuscular blockade:
  - 2 mg/kg of rocuronium IV, or
  - 2 mg/kg of suxamethonium IV.
- Intubate and immediately confirm ETT position with waveform capnography over the first seven breaths.
- Utilise the post intubation checklist.
- Administer ongoing sedation, analgesia, neuromuscular blockade and vasopressor using the **'post intubation'** guideline.

Adult dose summary table

	50 kg	60 kg	70 kg	80 kg	90 kg	≥ 100 kg
Fentanyl (2 mcg/kg)	100 mcg	120 mcg	140mcg	160mcg	180mcg	200mcg
Ketamine (1 mg/kg)	50 mg	60 mg	70 mg	80 mg	90 mg	100 mg
Ketamine (2 mg/kg)	100 mg	120 mg	140 mg	160 mg	180 mg	200 mg
Rocuronium	100 mg	100 mg	150 mg	150 mg	200 mg	200 mg
Suxamethonium	100 mg	100 mg	150 mg	150 mg	200 mg	200 mg

Paediatric dose summary table

	10 kg	20 kg	30 kg	40 kg
Fentanyl (2 mcg/kg)	20 mcg	40 mcg	60 mcg	80 mcg
Ketamine (1 mg/kg)	10 mg	20 mg	30 mg	40 mg
Ketamine (2 mg/kg)	20 mg	40 mg	60 mg	80 mg
Rocuronium	20 mg	40 mg	60 mg	80 mg
Suxamethonium	20 mg	40 mg	60 mg	80 mg

Additional information

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General principles

- The primary goal of performing RSI is to improve patient outcomes by securing the airway, treating hypoxia and controlling ventilation to prevent hypercarbia. This is particularly important in patients who are unconscious post cardiac arrest or have severe TBI, because secondary brain injury worsens long-term outcomes.
- Personnel calling for backup from a CCP must consider how long it will take for backup to arrive. To be appropriately utilised, such backup must arrive at least 15 minutes faster than the patient can be transported to an appropriate hospital.
- Including the clinical trajectory of the patient in the decision to perform RSI requires good clinical judgment. A clinical trajectory requiring RSI requires that the expectation is that the patient is likely to be admitted to an ICU ventilated, for example because the patient has severe TBI or is post cardiac arrest. A clinical trajectory that the patient may require intubation and ventilation for a procedure, such as surgery or a CT scan, is not usually considered an indication for RSI.
- Whenever feasible, an ICP or second CCP should be utilised as a dedicated assistant.
- If hypoxia is severe or the patient is at high risk of desaturation, consider oxygen via nasal prongs at 15 litres/minute during laryngoscopy.

PODCASTS acronym

- The decision to perform RSI requires clinical judgement that considers all factors contributing to the balance of risk for each patient. Use of the PODCASTS mnemonic provides a structure to help ensure key factors are considered. Personnel must have a low threshold for seeking clinical advice if unsure.

P Predicted difficulty of intubation. If the intubation is predicted to be difficult, the balance of risk usually shifts toward the RSI occurring in hospital.

O Oxygenation and ventilation (severity of impairment). If there is severe impairment of oxygenation or ventilation (particularly oxygenation) that cannot be managed with basic airway manoeuvres and/or adjuncts, the balance of risk usually shifts toward RSI occurring out of hospital.

D Distance (time) to hospital. If the time to the most appropriate hospital, including extrication time, is prolonged, the balance of risks usually shifts toward RSI occurring out of hospital. There is no time 'cut off', but it should be unusual to perform RSI within 15 minutes of the most appropriate hospital.

C Comorbidities that limit life expectancy. The presence of life limiting comorbidities usually shifts the balance of risk toward RSI occurring in hospital, because of the implications for ongoing treatment such as admission to an intensive care unit.

A Age. For those aged less than 5 years or greater than 75 years, the balance of risk usually shifts

toward RSI occurring in hospital.

S Shock. The presence of shock, particularly hypovolaemic shock, usually shifts the balance of risk toward RSI occurring in hospital. Positive pressure ventilation will further reduce venous return to the heart, and concurrent resuscitation of the patient's circulation, for example with blood, is usually required.

T Trajectory of clinical course. If the patient has an expected clinical trajectory of not improving rapidly, for example ROSC following prolonged cardiac arrest, the balance of risk usually shifts toward RSI occurring out of hospital. Conversely, a trajectory of expected improvement such as GHB poisoning usually shifts the balance of risk toward RSI occurring in hospital.

S Special circumstances. This includes all additional factors that alter the balance of risk. For example, scene factors such as availability of personnel to assist with the RSI, and specific patient factors such as morbid obesity and pregnancy.

Choice and dosing of medicines

- Fentanyl should usually be administered immediately prior to ketamine when performing RSI for patients with traumatic brain injury, suspected intracranial pathology (for example haemorrhagic stroke) or hypertension (a SBP greater than 150 mmHg in an adult or a SBP significantly higher than normal in a child).
  - This is because fentanyl may reduce the rise in BP and/or intracranial pressure, that may occur with laryngoscopy and intubation.
  - 2 mcg/kg (to a maximum of 200 mcg) is an appropriate dose for most patients.
  - However, fentanyl may contribute to hypotension. For this reason, fentanyl should be withheld if there are signs of shock.
- Ketamine is always administered during RSI.
  - 2 mg/kg (to a maximum of 200 mg) is an appropriate dose for most patients.
  - Although ketamine usually produces a small rise in BP, ketamine may cause a fall in BP if shock is present. For this reason, reduce the dose to 1 mg/kg if there are signs of shock.
  - Consider reducing the dose to 1 mg/kg if sedative medicines have been administered, or taken. For example, consider reducing the dose to 1 mg/kg if:
    - A sedative dose of ketamine has been administered as part of a delayed sequence intubation, or
    - The patient has taken sedative medicines.
  - Do not reduce the dose of ketamine in the setting of RSI for prolonged seizures. Even though midazolam will have been administered, an anticonvulsant dose of ketamine is required.
- Neuromuscular blockade is always administered during RSI.
  - 2 mg/kg of rocuronium or suxamethonium (to a maximum of 200 mg) is an appropriate dose for all patients.

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- The dose should usually be rounded to the nearest 50 mg in adults.

Delayed sequence intubation

- Delayed sequence intubation is a term used to describe the administration of sedation prior to embarking on the RSI procedure.
- Delayed sequence intubation is indicated if the patient meets criteria for RSI, but is agitated and/or combative to a level that is preventing safe preparation prior to RSI:
  - Administer sufficient ketamine (usually 1 mg/kg IV) to gain control of the agitation.
  - Once control is gained, ensure adequate pre-oxygenation and complete the RSI checklist.
  - Perform RSI and consider reducing the dose of ketamine, considering the dose of medicines already administered.

Crash RSI

- Rarely, a patient may be deteriorating so rapidly that there is a high risk of death occuring prior to performing RSI, and a truncated (or 'crash') approach should be undertaken.
- Task one person to provide as much as pre-oxygenation as possible.
- Utilise the crash RSI checklist.
- Omit fentanyl and administer ketamine and neuromuscular blockade.

Intubation without medicines

- For the majority of unconscious patients not in cardiac arrest, the risks of intubation without medicines outweigh the benefits.
- Intubation and ventilation without medicines must only occur if the patient has a GCS of 3 and ineffective breathing.

Measurement of end tidal CO_2 ($ETCO_2$)

- Intubation must not be attempted unless measurement of $ETCO_2$ via capnography is immediately available.
- Confirmation of correct placement of the ETT via capnography over the first seven breaths must occur as the first step following placement of an ETT. If the ETT is within the oesophagus and the stomach

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contains CO_2 (for example from drinks containing CO_2, or following mouth to mouth ventilation), $ETCO_2$ may be detectable for the first 2-4 breaths and then fall rapidly below 7 mmHg.

- Continuous measurement of the $ETCO_2$ via capnography is compulsory for all patients intubated with an ETT, including those in cardiac arrest.
- The ETT must be removed if the $ETCO_2$ is below 7 mmHg (including during cardiac arrest), even if it is thought that the low $ETCO_2$ is due to technical error. This is because the risk of unrecognised oesophageal intubation is too high, even when the ETT is thought clinically to be within the trachea.

More Information

Version history

28 October 2024 1.0.5
Update: Multiple changes.